



Search ...

Search where?

SEARCH

PROTEOMICS

PROTEIN-ID AND PROTEIN COVERAGE

For identification, sequence confirmation purposes and quality control we offer protein digestion from in-gel or in-solution samples.

For in-gel samples:

- **Coomassie-stained gels** We recommend to use home-made coomassie stains. Gels should be stained for the minimum time to only visualize your proteins of interest; extended staining will increase the background in the MS analysis.
- **Silverstained gels**
- Please provide us with the entire gel and a corresponding gel image indicating the bands that should be analyzed.

In-solution:

- Buffers should be detergent-free.
- Stoichiometries of proteins of interest in the sample can be estimated using iBAQ intensities.

In case you would like to have specific protein sequences of your interest included in the data analysis or if you have any tag attached to the original sequence (e.g. his tag, Flag, MBP, GST etc.) please send the **protein sequence in fasta format**.

PTMOMICS

We also offer the identification and/or quantification of post-translational modifications (PTM) of proteins e.g. mapping PTM sites in purified proteins or global mapping from complex samples.

- Phosphorylation, Ubiquitin, Acetylation, Methylation, Glycosylation...
- Low stoichiometry modifications can be brought into view by enrichment (e.g. Fe-IMAC for phosphorylation using an automated, highly reproducible platform).

EXPRESSION PROTEOMICS

"Total Proteomes"

In this approach, a quantitative comparison of total proteomes (or sub-proteomes) is performed to identify proteins that are differentially expressed in related samples/conditions (e.g. +/- drug treatment, diseased vs. healthy tissue, expression of proteins in different cell types).

- **Quantification:**
 - **label-free quantitation:** Samples are measured in triplicates at the very least (quadruplicates preferably) to derive statistically meaningful conclusions. Single sample measurements should only be used to validate an experimental set up or sample preparation protocol prior to submission of the final experiment.
 - Other quantification strategies that are possible with us:
 - Chemical labelling: **TMT labelling**
 - Metabolic labelling: **SILAC**
- For deep proteome profiling, we offer fractionation by offline SCX or highpH.

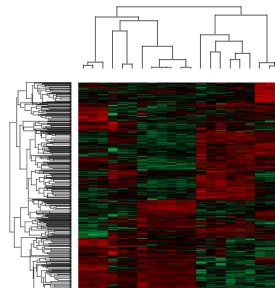


Figure 1 Heatmap of proteins identified in specific cells types showing the differential expression.

INTERACTION PROTEOMICS / AP-MS

"Interaction Proteomics"

For the identification of protein interaction partners, we commonly offer co-immunoprecipitation/co-affinity purification together with quantitative mass spectrometry (Figure 1).

- Samples can be provided as beads after a washing step for on-bead digestion, after elution for in-solution digestion (free of MS incompatible detergents) or loaded on a gel for in-gel digestion.
- For the label-free quantitation, samples need to be submitted in triplicates at the very least (quadruplicates preferably)

A: Affinity purification – mass spectrometry workflow

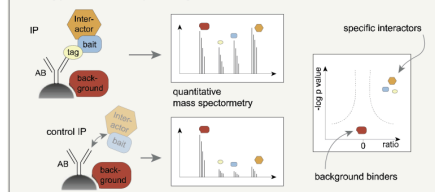


Figure 2 AP-MS workflow - The bait protein is pulled down together with interacting proteins, which are subsequently identified by mass spectrometry. A control sample not containing the bait serves as a control. True interactors can be distinguished from contaminants by extracting their respective abundances in bait-samples and control. While contaminants are equally present in both samples, specific interactors are enriched in the bait sample.

Last modified: 5/11/2021



Mass Spectrometry Core Facility

Contact



Barbara Agnes Steigenberger
STEIGENBERGER@BIOCHEM.MPG.DE
+49 89 8578-3799



Nicole Kromholz
KROMHOLZ@BIOCHEM.MPG.DE
+49 89 8578-3906



Anja Wehner
WEHNER@BIOCHEM.MPG.DE
+49 89 8578-3798

Email address



Mass Spectrometry
ms-service@biochem.mpg.de

Form



Proteomics

Form for proteomics measurement
04/19/2021 10:52



Mass Spectrometry Core Facility

Guidelines

Intact Mass

Proteomics

Crosslinking - MS





Assa Yeroslaviz



MY MPG

MY INSTITUTE

MY CAREER

SERVICE

COLLABORATION

ALL MPI / HQ

SEND →